

SYSTEM REQUIREMENTS DOCUMENT (SRD)

Coffee Shop Chain – Full Business Intelligence Project

1. Data Architecture

Model Type: Star Schema

Fact Table: FactSales

- Transaction ID
- Transaction Date
- Transaction Time
- Transaction QTY
- Unit Price
- Product ID
- Store ID

Dimension Tables

- DimStore (Store ID, Store Location)
- DimProduct (Product ID, Category, Type, Details)
- DateTable (Calendar table)
- DimTime (Hour, Minute, Time Period)

2. Data Dictionary

A full dictionary will be generated after dataset is uploaded. Initial definitions include:

Column	Type	Description
Transaction Date	Date	Date of purchase
Transaction Time	Time	Time of purchase
Transaction QTY	Integer	Quantity purchased
Unit Price	Decimal	Price per unit
Product Category	Text	High-level product grouping
Product Type	Text	Sub-category

Product Details	Text	Specific product name
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3. Required Transformations

- Convert Transaction Date to proper date type
- Extract hour from Transaction Time
- Create Revenue = Unit Price × Quantity
- Create Time Period (Morning, Afternoon, Evening)
- Build DateTable using CALENDAR()
- Build DimTime using generated timetable
- Ensure relationships: FactSales → DimStore, FactSales → DimProduct, FactSales → DimDate, FactSales → DimTime

4. Business Logic & DAX Requirements

Core Measures

- Total Sales
- Total Quantity
- Average Price
- Sales per Transaction
- Category Contribution %
- Store Ranking

Advanced DAX

- Rolling 30-day sales
- Rolling 12-month sales
- Month-over-Month %
- Year-over-Year %
- Dynamic Top N products
- Hour-of-day performance
- Time period segmentation
- Category mix analysis
- Virtual table calculations for ranking

5. Visual Requirements

Dashboard Pages

- Executive Summary
- Store Performance
- Product & Category Analysis
- Time Intelligence (Hour, Day, Month)
- Advanced Insights (Top N, Trends)

Key Visuals

- Line charts (trends)
- Bar charts (ranking)
- Heatmaps (hour × day)
- Donut charts (category mix)
- KPI cards
- Decomposition tree (optional)

6. User Interactions

- Slicers: Date, Store, Category, Product
- Drill-through: Store → Product
- Tooltips with advanced DAX
- Dynamic Top N selector

7. Performance Requirements

- All measures optimized using VAR + CALCULATE
- Avoid row-context iteration unless necessary
- Use Star Schema for maximum efficiency
- Limit visuals per page to maintain responsiveness

8. Security & Access

- Public portfolio project → no row-level security required
- No sensitive data included